

9(a)(i)	any three from: 1 (random) mutation(s) allows detection of high frequency ; 2 selection pressure is bat (predation) ; 3 those that can detect, high frequencies / bats, survive / don't get eaten / have selective advantage / are selected for ; 4 these, reproduce / pass on (beneficial) allele(s) ; 5 those that can detect high sounds / beneficial alleles, increase (in frequency / in population) ;	3
9(a)(ii)	any four from: 1 <i>ref. to</i> receptor / generator, potential ; 2 vesicles of (named) neurotransmitter, move towards / fuse with, (presynaptic) membrane ; 3 (named) neurotransmitter, release / exocytosis ; 4 neurotransmitter binds to receptors on sensory neurone ; 5 Na⁺ / sodium, ions enter (sensory neurone) ; 6 depolarisation (of membrane of sensory neurone) ; 7 <i>ref. to</i> threshold potential / all-or-nothing response ;	4

Question	Answer	Marks
9(b)	any three from: S1 sympatric (speciation) ; S2 different mutations ; S3 gave, (named) phenotypic / behavioural, differences ; S4 which gave behavioural, isolation / separation ; S5 (leading to) inability to reproduce together / reproductive isolation / genetic isolation / no gene flow between them ; or any three from: A1 allopatric (speciation) ; A2 geographical, separation / barrier, in the past ; A3 different, mutations / selection pressures ; A4 gave, (named) phenotypic / behavioural, differences ; A5 inability to reproduce together / reproductive isolation / genetic isolation / no gene flow between them, after populations re-join ;	3